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Can AI Detect Tail Events? Stock Performance Around Tail Events During Different Sentiment Regimes

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ABSTRACT

This study examines the dynamics of US stock-specific news sentiment relative to aggregate market sentiment, leveraging a dataset of 2.7 million news articles. Focusing on negative tail events among S&P 100 constituents from 2003 to 2021, we identify a significant association between firm-specific sentiment changes and subsequent one-month stock performance. Specifically, a positive news sentiment score during adverse events suggests the absence of fundamental issues and correlates with superior post-event returns. These findings support the development of a profitable reversal trading strategy and indicate a potential regime shift in market behavior.

KEYWORDS

Behavioral Finance; Efficient Market Hypothesis; News Sentiment; Overreaction; Tail Events; Underreaction

JEL CODES

G11; G40

1. Introduction

The stock market is driven by behavioral factors and shows excessive volatility and overreaction to new information (Shiller 1981). Furthermore, there is an overreaction to positive firm news with a reversal and an underreaction to negative company news with a further downwards drift. However, stock sell-offs accompanied with no bad firm-specific news tend to reverse subsequently (Frank and Sanati 2018). These behavioral effects are even more pronounced in times of high uncertainty on the market's future outlook. There were several shocks in the recent stock market development, such as the financial crisis or the Covid-19 pandemic. During such periods with high uncertainty and excess volatility, panic and fear led to even higher overreactions. Such market circumstances often trigger significant market movements and even larger individual stock price jumps. This can also be observed by the distribution of the daily stock returns, which is not normally distributed as often assumed, but rather has fat tails on both ends of the return spectrum (Brown and Warner 1985).

Focusing specifically on tail events in the daily stock return distribution, the importance of such big return jumps becomes apparent. Missing the best (worst) days would significantly decrease (increase) the long-term performance of the underlying asset. For example, between 2001 and 2020, missing the best ten days of the S&P 500

market index would cut the performance in half. Moreover, the worst performing days are often followed by the best performing days. Seven out of the ten best performing days occur on average within the following two weeks of the ten worst days. Such short-term price overreactions are mostly driven by psychological factors rather than fundamental information (Morgan 2021).

In most cases, these rare negative market shocks are unpredictable and, therefore, hard to prevent. Because extreme return days have an enormous impact on the creation of long-term wealth, we mainly focus on such rare tail events. Based on the black swan metaphor of Taleb (2007), such extreme events are very exceptional and not predictable. However, from an ex-post perspective after the occurrence of such price drops, it could be insightful to develop a decision criterion to assess the extent of a stock price recovery in the following time period. Such an analysis might allow to distinguish between temporary and ongoing stock sell-offs and hence also identify cases where it is beneficial to “buy the dip” or “catch a falling knife”. Such a measure could differentiate between market co-movements and stock-specific movements caused by firm-specific fundamental information. If the stock price decline is caused by fundamental news of the company itself, it might indicate a firm-specific issue with no reversal or even a further stock price decline. On the other hand, if the price

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drop is a result of the overall negative market circumstances, this might indicate no fundamental issue of the company. Consequently, there might be a price reversal in the subsequent period after the big stock sell-off.

To assess the post-event performance of the S&P 100 constituents, we construct a decision criterion based on news sentiment data. The ultimate goal of this measure is to identify firms that have experienced a negative tail event with no fundamental problem within the company that might have just suffered a shock based on the overall negative market circumstances. Such an evaluation would help to distinguish between temporary and ongoing stock sell-offs. By doing so, we shed some light on the underlying research question: *What power does news sentiment have in explaining if and to what extent companies recover from negative tail events?*

In order to examine this research question, we construct a relative stock-specific news sentiment indicator, which captures the difference between the individual stock and the overall market news sentiment change. By doing so, we are able to quantify the news tone change of each firm relative to the market. Derived from the research question, we investigate the following hypothesis: *Companies that experience a negative tail event during negative market days accompanied with a positive relative stock-specific news sentiment change have on average a higher post-event performance compared to companies with a negative one.* Put differently, the extent of a tail event reversal is conditional on the relative firm-specific news sentiment change. Consequently, companies with a positive news sentiment measure might exhibit a higher post-event return compared to companies with a negative one. A positive relative stock-specific news sentiment change could be a sign that the negative tail event was mainly caused by the overall market circumstances without a firm-specific fundamental problem, which may result in a post-event reversal. In the opposite case, a negative relative stock-specific news sentiment change might indicate a fundamental issue of the specific company, which could experience a further downside drift, or may result in bankruptcy in the worst case.

Previous studies mainly emphasize the existence or quality of news articles near tail events. In order to go a step further and to examine the underlying research question, we focus on the S&P 100 constituents between 2003 and 2021. Since we are mainly interested in tail events, we define jumps based on the stock return distribution and only utilize the events at the left tail of the distribution. Furthermore, we aggregate the article-based Reuters news sentiment data to

a daily relative stock-specific news sentiment level for each company by subtracting the overall market news tone. To use a predictive news sentiment measure, the analysis is based upon the news sentiment change between the 1-month and 12-month moving average (a common price momentum frequency). We employ this momentum measure to aggregate the limited daily stock-specific news coverage and to filter the noisy news sentiment data. Because the tail events differ across firms and time, we utilize a fixed effects regression setting. In order to assess the hypothesis, we study the relationship between the relative change in company news tone and the performance of one month after the tail events. Based on the statistical results, we construct an implementable reversal trading strategy, which invests conditional on the created relative news sentiment measure.

With the relative stock-specific news sentiment change, we introduce a measure that helps to explain the extent of the recovery one month after tail events of S&P 100 constituents. During the sample period between 2004 and 2021, firms that have a positive relative stock-specific news sentiment change at the tail event, experience on average a significantly higher post-event performance over the following month compared to firms with a negative sentiment measure. Based on the statistical insights, we are able to present a profitable conditional reversal trading strategy that outperforms the S&P 100 index. The strategy invests in stocks that had a negative tail event during negative market days accompanied by a positive news sentiment measure. The positive news sentiment measure signals no company-specific fundamental issue within the firm itself. This suggests that the negative stock jump is an overreaction caused by the negative overall market circumstances, which exhibits an abnormal reversal over the following month.

In summary, our findings suggest that information now moves from news media to financial markets more quickly than before. This faster pace may result from increased news reporting, a higher concentration of investors who respond to news, and improvements in automated systems that process news articles. Additionally, changes in how markets operate—particularly the growing role and influence of high-frequency trading firms—may have further strengthened this trend.

2. Literature survey

Antweiler and Frank (2006) developed a sentiment analysis algorithm using Wall Street Journal articles to challenge the Efficient Market Hypothesis, finding

evidence of market reversals and overreactions to news, with more pronounced and lasting effects during recessions. Tetlock (2007) advanced quantitative news analysis with a bag-of-words method and the Harvard dictionary, revealing that negative news sentiment predicts short-term market downturns followed by quick reversions to fundamentals. De Bondt and Thaler (1985) showed that dramatic, unexpected news leads to stock price overreactions, though not necessarily irrational market behavior. Cutler, Poterba, and Summers (1989) found that some price movements occur without major news, suggesting news only partially drives market changes.

Veronesi (1999) demonstrated that market sensitivity to news depends on economic conditions, with overreactions to negative news in good times and underreactions during downturns. Barberis, Shleifer, and Vishny (1998) noted overreactions during crashes, even without significant fundamental news. Baker and Wurgler (2007) showed that sentiment becomes especially influential during market distress, leading to misvaluation. García (2013) found that news sentiment from New York Times articles can forecast daily DJIA returns, particularly during recessions, highlighting the market's economic context dependence. Baker, Bloom, and Davis (2016) created an economic policy uncertainty index based on news coverage frequency, which spikes during major events and correlates with higher market volatility. Manela and Moreira (2017) constructed an uncertainty index from WSJ front-page news, showing its highest values during crises and its predictive power for future returns in stable periods.

Baker et al. (2021) analyzed international market jumps, finding that downward moves linked to non-policy news are twice as common as upward ones, and policy news often triggers positive jumps. Market volatility is higher after jumps tied to unclear causes. Chan (2003) examined momentum and reversal after price jumps with and without news, identifying long-term drifts after bad news and short-term reversals when no public information was present, consistent with investor underreaction and later overreaction.

Pritamani and Singal (2001) observed that post-event returns depend on the quality of news, with earnings and analyst-related news driving significant price continuation. Savor (2012) showed price shocks accompanied by information result in drift, while those without evidence lead to reversals, especially for negative shocks. Market participants underreact to fundamental news but overreact to external factors, supporting trading strategies that exploit these patterns. Boudoukh et al. (2013) found reversals after

extreme return days lacking fundamental news, while continuation followed days with news. Jiang and Yao (2013) noted that small, value, or illiquid stocks have larger price jumps, and past losers tend to outperform winners after jumps.

Frank and Sanati (2018) highlighted that positive price jumps are followed by reversals, while negative jumps see continued declines, unless there is no identifiable bad news, in which case reversals occur. Allen, McAleer, and Singh (2019) showed news sentiment has a stronger impact on negative extremes of stock returns for DJIA constituents. While many studies connect return jumps with news sentiment, few examine a stock-specific sentiment measure relative to the overall market for predicting reversals after negative tail events.

3. Empirical analysis

3.1. Background

The overreaction hypothesis contributes to behavioral biases in financial markets. By analyzing negative tail events using stock returns and news sentiment, we examine excess volatility and assess post-shock performance.

3.1.1. Idea and intuition

The concept for this analysis was developed during the Covid-19 pandemic, a period marked by a significant increase in market uncertainty. Due to government restrictions and an adverse macroeconomic environment, many stocks experienced substantial price declines in February and March 2020, even though there were no fundamental changes within many companies. The interconnectedness of financial markets contributed to widespread negative co-movement, affecting companies with otherwise positive outlooks. During periods of volatility, strong market recoveries can follow declines, as observed with the rebound of the stock market beginning in late March 2020. The unpredictability of the Covid-19 downturn made it challenging to anticipate or prevent losses. From a retrospective viewpoint, it may be useful to develop a decision criterion to assess the likelihood of firm-specific recoveries following significant negative events. This analysis employs article-based news sentiment data to create both stock-specific and overall market sentiment indicators. Such measures could help differentiate between temporary and persistent stock sell-offs, and may assist in identifying situations where buying opportunities exist after a sharp decline. Ideally, the criterion would distinguish between

market-wide movements and company-specific developments. If a stock's decline is due to internal, fundamental changes, it may signal ongoing challenges without expected reversal. Conversely, if the decline results from broader market conditions, it could suggest the absence of company-specific issues and the potential for a subsequent price recovery.

3.1.2. Research question and hypothesis

To assess the stock performance after negative tail events, we combine stock prices with news sentiment data to assess the following research question: *What power does news sentiment have in explaining if and to what extent companies recover from negative tail events?*

To analyze this research question, we construct a relative stock-specific news sentiment change, which is similar to the methodology of Uhl and Novacek (2021). The relative measure captures the difference between the individual stock and the overall market news sentiment change. By doing so, we are able to quantify the news tone change of each firm relative to the market, which enables us to investigate the following hypothesis: *Companies that experience a negative tail event during negative market days accompanied with a positive relative stock-specific news sentiment change have on average a higher post-event performance compared to companies with a negative one.* Put differently, the extent of a tail event reversal is conditional on the relative firm-specific news sentiment change. Hence, companies with a positive news sentiment measure might exhibit a higher reversal compared to companies with a negative one. The positive relative stock-specific news sentiment change could be a sign that the tail event was mainly caused by the overall market circumstances without a firm-specific fundamental problem, which might result in a post-event reversal. In the second case, the negative relative stock-specific news sentiment change might indicate a fundamental issue within the specific company that might cause a further downside drift or in the worst case might result in bankruptcy.

3.1.3. Methodology

To test our hypothesis, we analyze stock market and news sentiment data for S&P 100 companies from 2003 to 2021. Tail events are defined as extreme returns using a 252-day rolling window. We assess changes in stock-specific news sentiment—derived from 2.7 million Reuters articles—relative to all S&P 100 firms. Fixed effects models control for firm- and time-specific factors. Using these results, we develop a

trading strategy based on the news sentiment measure, applying it both in- and out-of-sample.

3.2. Data

This analysis examines the US market from January 2003 to December 2021. Using Bloomberg stock data and Thomson Reuters News Analytics (TRNA) news sentiment scores, we study daily returns following company-specific tail events. Due to coverage limits, only S&P 100 firms are included, with news sentiment aggregated daily.

3.2.1. Stock and market data

Financial data pertaining to the S&P 100 index—including its constituents at specific points in time and the risk-free rate between 2003 and 2021—were obtained from Bloomberg. Daily returns were calculated based on closing prices. Across the 19-year study period, data were available for 177 S&P 100 constituents. Restricting the sample to active constituents at each point in time ensures that data for 96 to 100 companies are present on any given trading day (See Table A2 in the Appendix for the full company universe covered in the study).

In addition to calculating normal stock returns, abnormal returns were computed by subtracting the daily S&P 100 index return from individual stock returns. This approach facilitates analysis and interpretation of company performance relative to the broader US market.

To identify stock price jumps or tail events within the return time series, a dynamic return distribution was constructed for each stock and for the overall S&P 100 index on a daily basis. The distribution's mean and standard deviation were estimated using a rolling look-back window of the preceding 252 trading days. As a result, tail events could be identified from January 2004 onward. This methodology closely follows Pritamani and Singal (2001), who argue that an absolute percentage threshold—used in prior studies such as Atkins and Dyl (1990) and Bremer and Sweeney (1991)—can introduce bias by disproportionately selecting volatile stocks. The adopted approach accounts for stock-specific volatility, the evolving variability of returns over time, and helps to prevent look-ahead bias.

Assuming normally distributed returns, approximately 95.5% (99.7%) of daily returns should fall within ± 2 (± 3) standard deviations of the mean. This implies that, out of 252 trading days, 11.34 (0.756) days would theoretically fall outside the 95.5% (99.7%)

Table 1. Yearly number of stock-specific tail events.

Year	Positive tail events (2 std)	Negative tail events (2 std)	Positive tail events (3 std)	Negative tail events (3 std)
2003	—	—	—	—
2004	425	363	105	78
2005	539	490	142	99
2006	672	484	170	115
2007	829	1'081	228	289
2008	1'566	1'774	610	600
2009	303	180	75	25
2010	440	424	72	65
2011	886	1'043	220	351
2012	234	244	53	60
2013	474	456	114	105
2014	554	719	122	155
2015	778	893	202	246
2016	511	555	126	166
2017	405	416	117	132
2018	787	1'289	174	435
2019	385	488	93	107
2020	1'051	1'225	540	638
2021	246	304	51	77
Mean	583	654	169	197
Total	11,085	12,428	3,214	3,743

The table shows the yearly total number of positive and negative stock return jumps of the active S&P 100 constituents over the examined period. The tail events are based on the rolling mean and standard deviation with a look-back window of 252 trading days. A similar table for the overall market is provided in [Table A1](#) the appendix.

interval, corresponding to 5.67 (0.378) positive or negative tail event days per side annually.

Using the rolling window methodology, both positive and negative tail event signals were identified for each company and for the S&P 100 market as a whole. [Table 1](#) reports the annual number of stock-specific tail events, demonstrating that negative tail events generally occur more frequently than positive ones. Additionally, there is considerable variation in the frequency of stock price jumps over time, with notable increases observed during the financial crisis and the COVID-19 pandemic; similar trends are evident at the market level.

According to Taleb's Taleb (2007) definition, events falling beyond three standard deviations may be classified as rare and unpredictable "black swan" occurrences. However, applying this threshold did not yield a sufficient number of tail event signals for the development of practical trading strategies. Consequently, a tail event in this study is defined as a return that falls outside the ± 2 standard deviation interval from the mean. Emphasis is placed primarily on negative stock jumps, focusing on the left tail of the return distribution.

3.2.2. News sentiment data

Refinitiv News Analytics (TRNA) applies a financial-domain NLP framework that combines dictionary-based sentiment classification with contextual parsing and relevance weighting to assign firm-level sentiment scores to individual news items. While the underlying algorithms are proprietary, the methodology and

empirical properties of the data are extensively documented (Refinitiv 2019) and widely validated in the academic literature. S&P 100 news sentiment data comes from the TRNA database, which provides article-based sentiment and metadata dating back to 2003. Each entry includes a timestamp, topic codes, company IDs, asset names, and a sentiment score (positive, neutral, or negative). Sentiment is determined by a blend of rule-based and deep learning approaches using NLP. [Table 2](#) lists five recent sample articles with their metadata.

We analyze news articles for North American companies in the S&P 100 between 2003 and 2021, filtering using Thomson Reuters PermIDs to ensure consistent identification of index members. Only articles published while a company is part of the S&P 100 are included; for instance, Tesla's news is considered from December 21, 2020, onwards, while Occidental Petroleum's is included only until its removal on December 20, 2020. [Figure 1](#) shows that daily news volume for S&P 100 constituents rose steadily from 2003 to 2008, with spikes around quarterly results. Following Uhl, Pedersen, and Malitius (2015), we focus exclusively on trading days, as week-end news is sparse and often repeats earlier events.

After filtering, the dataset includes 2.7 million Reuters articles mentioning S&P 100 firms from 2003 to 2021. [Table 3](#) presents yearly news sentiment data, showing that average sentiment fluctuates but is generally negative, notably during the financial crisis. Based on TRNA sentiment classes, negative news articles consistently outnumber positive ones.

Table 2. TRNA data structure of selected columns.

feedTimestamp	sentimentClass	subjects	assetCodes	assetName	headline
2021-12-31 17:57:39.826	1	A:1 A:C G:6J M:NY M:Z B:164 B:1740 M:2DZ G:4 G ...	P:4295914405 R:NVDA.O R:NVDA.OQ	NVIDIA Corp	BUZZ-Nvidia powered through 2021, other chipma ...
2021-12-31 18:32:56.988	-1	A:1 A:C G:6J M:NY M:Z B:173 B:174 B:1800 B:180 ...	P:4297297477 R:FB.O R:FB.OQ	Meta Platforms Inc	BUZZ-After bracing a turbulent year, Meta Plat ...
2021-12-31 19:09:20.81	-1	B:261 E:K G:6J M:NY M:Z M:E7 A:5 B:128 B:1615 ...	P:8589934339 R:BAC.N R:BAC pb.N R:MER -pk.N	Bank of America Corp	BofA allows remote work in early January source
2021-12-31 21:47:19.341	-1	B:1348 B:200 B:255 B:256 B:258 B:69 B:75 B:81 ...	P:4295904620 R:NKE.DE R:NKE.N	Nike Inc	Refinitiv Newscasts Actress Betty White dies ...
2021-12-31 23:20:43.066	-1	B:1137 B:1266 B:177 B:178 B:179 B:1811 B:1817 ...	P:4295904853 R:T.DE R:T.N	AT&T Inc	U.S. officials ask AT&T, Verizon to delay 5 G w ...

The table illustrates five recently published articles on December 31, 2021. The first five reported columns are the basis for the news sentiment aggregation of the analysis. This selection is only a fraction of the provided TRNA metadata.

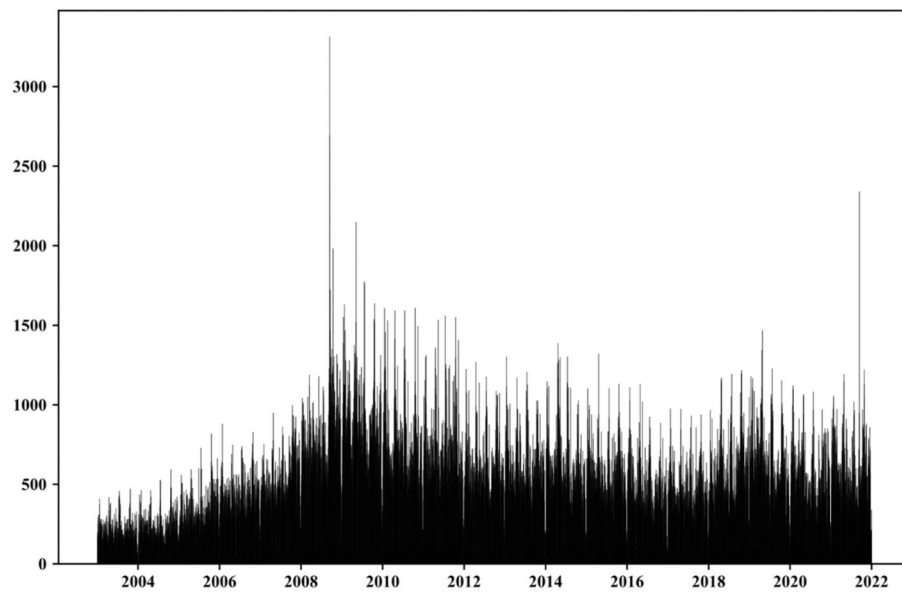


Figure 1. Daily news volume of S&P 100 constituents. The figure illustrates the total daily news volume (*y-axis*) for the active S&P 100 constituents on each day over the examined period (*x-axis*) from January 1, 2003 to December 31, 2021.

Since few new articles are published daily for S&P 100 constituents, sentiment analysis uses a forward fill method, replacing missing values with the previous day's sentiment (Coqueret 2020). This approach ensures sentiment only changes when new articles appear. To minimize noise, rolling moving averages (MA) aggregate article sentiment over weekly to yearly periods. Each article is equally weighted by dividing the total sentiment by the number of articles, accommodating daily variance, which avoids introducing arbitrary bias when no article has a clearly superior informational value.

A positive relative stock-specific sentiment change does not imply that firm-level news dominates market conditions, but rather that firm-specific information is comparatively benign, allowing market-driven price pressure to reverse once aggregate sentiment stabilizes. This aggregation filters out extreme sentiment fluctuations, similar to Uhl and Novacek (2021), and applies both at the company and market level—market sentiment is constructed from the same firm-level news, not macroeconomic sources. Relative sentiment for each firm is calculated by subtracting market

Table 3. Yearly news sentiment.

Year	Yearly news volume	Mean news sentiment	News volume positive (%)	News volume neutral (%)	News volume negative (%)
2003	54,701	−0.128	34.577	18.062	47.361
2004	61,145	−0.038	37.128	21.902	40.970
2005	86,088	0.004	34.023	32.384	33.592
2006	110,072	0.019	32.277	37.388	30.335
2007	136,344	−0.014	30.434	37.766	31.800
2008	207,397	−0.139	25.551	35.011	39.439
2009	222,445	−0.062	29.945	33.879	36.176
2010	187,953	−0.012	34.499	29.788	35.713
2011	180,471	−0.045	35.877	23.725	40.399
2012	151,330	−0.040	37.313	21.401	41.286
2013	154,862	0.028	39.633	23.557	36.810
2014	154,524	0.028	39.433	23.913	36.654
2015	143,915	0.006	37.867	24.886	37.247
2016	122,782	−0.004	39.709	20.216	40.075
2017	119,522	0.047	38.549	27.626	33.826
2018	147,668	−0.030	38.463	20.086	41.450
2019	162,002	−0.051	36.046	22.791	41.163
2020	137,605	−0.048	38.494	18.206	43.300
2021	152,922	−0.033	38.670	19.326	42.004
Mean	141,776	−0.027	35.710	25.890	38.400

The table summarizes the yearly aggregated news sentiment data for the active S&P 100 constituents over the examined period. It reveals the total yearly news volume, the yearly news sentiment mean, and the relative news volume for each sentiment class. The sentiment class is based on the TRNA sentiment classification method.

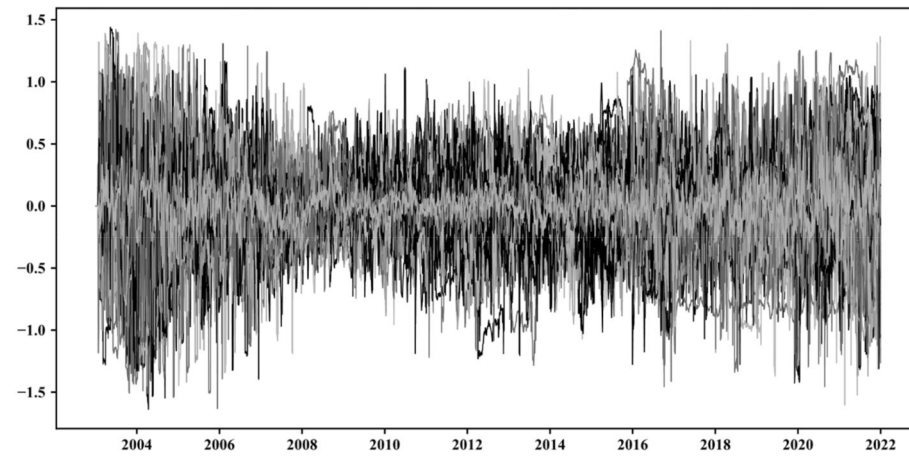


Figure 2. Relative stock-specific news sentiment 1-month over 12-month CMA. The figure shows the relative stock-specific news sentiment 1-month over 12-month CMAs (y axis) for the S&P 100 constituents over the examined period (x-axis) from January 1, 2003 to December 31, 2021. Taking the difference between stock-specific and market news sentiment results in a wider relative news sentiment range $[-2, +2]$. The graph only serves as guidance and is without a meaningful interpretation.

sentiment from firm sentiment, linking it to relative performance in the US economy.

Prior research (Uhl and Novacek 2021) shows that changes in news sentiment, rather than its level, better predict stock returns. Sentiment change is measured by crossing moving averages (CMA), such as comparing 1-month to 12-month MAs. Because individual firm news is sparse and noisy, a 1-month over 12-month CMA is used, as shown in Figure 2, allowing sentiment measures beyond class limits (± 1). While firm-specific sentiment remains somewhat volatile over 19 years, short-term CMAs are preferred to capture rapid shifts tied to stock jumps.

3.3. Statistical analysis

This section examines the research question regarding the relationship between relative stock-specific news sentiment change and post-event performance following negative tail events. The hypothesis under review is: *Companies experiencing a negative tail event during negative market days, along with a positive relative stock-specific news sentiment change, have, on average, higher post-event performance compared to those with a negative sentiment change.* This suggests that post-event reversals may occur following negative tail events, conditional on sentiment changes related to the company. The following null and alternative hypotheses are outlined:

Table 4. Tail event data structure (in-sample period).

Dates	PermID	Post-event CAR (20 days)	News sentiment CMA	Tail event return	Pre-event CAR (20 days)
2014-11-12	4295903216	1.018	0.030	−0.043	1.006
2014-11-17	4295904116	0.903	−0.201	−0.106	0.972
2014-11-25	4295903051	1.065	0.049	−0.022	1.014
2014-11-25	4295903449	0.986	−0.200	−0.032	0.988
2014-11-25	4295904888	1.007	−0.343	−0.033	1.002

The table shows the last five stock-specific tail events of the in-sample period between January 1, 2004 until December 31, 2014. It illustrates the underlying data structure used for the statistical analysis.

Table 5. Summary statistics (in-sample period).

	N	Mean	Std	Min	1. Quartile	3. Quartile	Max
Post-event CAR (20 days)	5,161	1.005	0.101	0.255	0.963	1.043	3.014
News sentiment CMA	5,161	−0.010	0.217	−1.508	−0.130	0.118	1.319
Tail event return	5,161	−0.051	0.037	−0.608	−0.059	−0.030	−0.011
Pre-event CAR (20 days)	5,161	0.991	0.091	0.181	0.955	1.034	2.020

The table shows a summary statistics of the four main variables used in the statistical analysis for the in-sample period between January 1, 2004 until December 31, 2014.

H0: There is no significant difference in post-event performance after negative tail events between positive and negative relative stock-specific news sentiment changes.

H1: There is a significant difference in post-event performance after negative tail events between positive and negative relative stock-specific news sentiment changes.

The hypothesis relates to investor reaction patterns such as underreaction and overreaction following negative tail events and subsequent positive reversal or negative drift. To distinguish between these cases, the analysis utilizes pre-processed news sentiment data from TRNA and stock market data from Bloomberg. Unique firm identifiers, including Thomson Reuters PermID and Bloomberg tickers, are used to match these sources. During the period from 2003 to 2021, 160 S&P 100 constituents have available data from both sources.

For development of a trading strategy informed by statistical analysis, the overall time span is divided into in-sample and out-of-sample periods. Statistical examination employs the in-sample period, covering ten years from January 1, 2004 to December 31, 2014. Within this interval, data is available for between 84 and 98 active S&P 100 constituents on each trading day, with the average number gradually increasing over time. The subsequent seven years, from January 1, 2015 to December 31, 2021, serve exclusively as the out-of-sample period for trading strategy evaluation, comprising data from 98 to 100 active constituents per trading day. The increase in examined constituents is primarily attributed to expanded news coverage.

The analysis focuses on negative tail events defined by a dynamic return distribution using a rolling look-back period of 252 days. Negative jump returns are

identified as those smaller than the mean return minus two standard deviations, or below 97.5% of other daily returns, assuming a Gaussian distribution. To incorporate market co-movement effects, the tail event signal is only considered if the market also shows a negative return on the same day. This approach accounts for broader economic influences on individual stocks, which may affect the likelihood of return reversals. Across the in-sample period, there are 5,161 negative stock-specific tail events among active constituents when the S&P 100 index also exhibited negative returns, all of which are included in the statistical analysis.

For measuring relative stock-specific news sentiment around tail events, the analysis uses the 1-month over 12-month CMA to capture changes in news tone surrounding significant stock movements. In line with previous studies (e.g., Pritamani and Singal 2001; Chan 2003; Savor 2012), an investment horizon of 20 trading days post-event is evaluated. To compare individual stock performance relative to the US market, post-event cumulative abnormal return (CAR) is calculated for the month following the event. A two-day trading lag ($t = [2, 22]$) is applied to the post-event CAR, consistent with Uhl and Novacek (2021). The negative tail event return is recorded for the event day ($t = 0$), and a pre-event CAR over the 20 trading days before the negative jump is measured up to the day prior ($t = [-21, -1]$). Table 4 presents the structure of the dataset, with each observation representing a stock-specific tail event on a particular day, including post-event CAR, news sentiment CMA, event day return, and pre-event CAR.

Table 5 presents the distribution of the main variables, including mean, standard deviation, minimum,

Table 6. Fixed effects regression (training period).

	Dependent variable Post-event CAR (20 days)		
	Default SE	Newey-West SE	Two-way cluster-robust SE
Tail event return	−0.458*** (0.059)	−0.458*** (0.174)	−0.458** (0.187)
Pre-event CAR (20 days)	−0.159*** (0.018)	−0.159** (0.072)	−0.159** (0.074)
Observations	5'161	5'161	5'161
R2	0.038	0.038	0.038
Adjusted R ²	−0.161***	−0.161	−0.161
F Statistic (df = 3; 4275)	56.067	56.067***	56.067***

The table shows fixed effects regressions of negative tail events of the S&P 100 constituents throughout the training period from January 1, 2004 to December 31, 2014. The CARs are calculated over 20 trading days before and after the tail event with a lag of two days. The regressions include default, Newey-West, and two-way cluster-robust SE.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

first quartile, third quartile, and maximum values for the 5,161 negative tail events observed during the in-sample period.

Drawing on time-series data for S&P 100 constituents and their respective stock-specific tail events occurring at varying times, we employ a fixed effects regression framework. The model utilizes the 1-month over 12-month relative, stock-specific news sentiment CMA for company i at time $t=0$ (*News sentiment CMA_i, $t=0$*) to explain the post-event cumulative abnormal return (CAR) over 20 trading days with a two-day lag (*Post-event CAR_i, $t=[2,22]$*). Controls include the tail event return at time $t=0$ (*Tail event return_i, $t=0$*) and the pre-event CAR covering the 20 trading days prior to the event (*Pre-event CAR_i, $t=[-21,-1]$*). To account for heterogeneity across firms and temporal variation, the specification incorporates time and firm fixed effects (X_k, it), firm-level intercepts (α_i), and an error term ($\epsilon_{i,t}$). The model is specified in Equation 1:

$$\begin{aligned} \text{Post-event CAR}_{i,t=[2,22]} &= \beta_{1_News\text{ sentiment CMA}_i, t=0} \\ &+ \beta_{2_Tail\text{ event return}_{i,t=0}} + \beta_{3_Pre-event\text{ CAR}_{i,t=[-21,-1]}} \\ &+ \beta_k X_{k,it} + \alpha_i + \epsilon_{i,t} \end{aligned} \quad (1)$$

This approach facilitates causal estimation of the post-event CAR while controlling for firm- and time-invariant factors. To mitigate autocorrelation and ensure robustness, we present fixed effects regression results using default, Newey-West, and two-way cluster-robust standard errors (SE) in Table 6. Results indicate that the coefficient on news sentiment CMA is positive and statistically significant at the 10% level, suggesting that, within firms and controlling for time, higher news sentiment CMA corresponds to increased post-event CAR over a 20-day holding period. Additionally, there is a negative association between tail event return and post-event CAR, indicating that larger tail events are generally followed by higher

CARs—reflecting a stronger post-event reversal when other factors are held constant. The model also identifies a negative relationship between pre-event and post-event CARs, signifying that firms experiencing a negative tail event with a higher pre-event 20-day CAR tend, on average, to exhibit lower post-event CARs, conditional on other variables. These outcomes lend support to the extensively studied overreaction hypothesis.

We use fixed effects regression, splitting the news sentiment CMA coefficient into three groups and four quartiles to test whether post-event CAR depends on sentiment categories. Both models use the lowest group as the benchmark and apply two-way cluster-robust standard errors. Equation 2 includes sentiment classes as dummies, referencing negative sentiment, while Equation 3 uses sentiment quartiles with the lowest quartile as the base case.

$$\begin{aligned} \text{Post-event CAR}_{i,t=[2,22]} &= \beta_1 \text{News sentiment CMA neutral}_{i,t=0} + \\ &+ \beta_2 \text{News sentiment CMA positive}_{i,t=0} + \\ &+ \beta_3 \text{Tail event return}_{i,t=0} + \\ &+ \beta_4 \text{Pre-event CAR}_{i,t=[-21,-1]} + \beta_k X_{k,it} + \alpha_i + \epsilon_{i,t} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{Post-event CAR}_{i,t=[2,22]} &= \beta_1 \text{News sentiment CMA quartile2}_{i,t=0} + \\ &+ \beta_2 \text{News sentiment CMA quartile3}_{i,t=0} + \\ &+ \beta_3 \text{News sentiment CMA quartile4}_{i,t=0} + \\ &+ \beta_4 \text{Tail event return}_{i,t=0} + \beta_5 \text{Pre-event CAR}_{i,t=[-21,-1]} \\ &+ \beta_k X_{k,it} + \alpha_i + \epsilon_{i,t} \end{aligned} \quad (3)$$

Table 7 presents two models with sentiment group dummy variables. The first model finds the positive sentiment group has a significantly higher ex-post CAR than the negative group, controlling for other factors. This supports focusing investments on firms with positive stock-specific news sentiment at the tail event. In the second model, splitting data into quartiles yields no significant differences from the base case, though coefficients rise with higher sentiment.

Table 7. Fixed effects regression news sentiment dummies (training period).

	Dependent variable	
	Post-event CAR (20 days)	
	Two-way cluster-robust SE (1)	Two-way cluster-robust SE (2)
News sentiment CMA neutral	−0.022 (0.020)	
News sentiment CMA positive	0.008** (0.004)	
News sentiment quartile 2		0.001 (0.004)
News sentiment quartile 3		0.003 (0.004)
News sentiment quartile 4		0.006 (0.004)
Tail event return	−0.458** (0.187)	−0.457** (0.187)
Pre-event CAR (20 days)	−0.158** (0.073)	−0.153** (0.072)
Observations	5'161	5'161
R ²	0.038	0.037
Adjusted R ²	−0.161	−0.163
F Statistic	42.259*** (df = 4; 4274)	32.911*** (df = 5; 4273)

The table shows fixed effects regressions with news sentiment dummy variables of negative tail events of the S&P 100 constituents throughout the in-sample period from January 1, 2004 to December 31, 2014. The CARs are calculated over 20 trading days before and after the tail event with a lag of two days. The regressions include two-way cluster-robust SE for time and firm. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

These results show that only the distinction between positive and negative sentiment groups is meaningful for the trading strategy.

The findings indicate that the null hypothesis—There is no significant difference in post-event performance after negative tail events between a positive and a negative relative stock-specific news sentiment change—can be rejected. Accordingly, the analysis demonstrates that the constructed relative stock-specific news sentiment CMA provides valuable insight for determining whether to “catch a falling knife” or “buy the dip” following negative return jumps during the training period from 2004 to 2014.

4. Trading strategy

4.1. Idea and intuition

The empirical analysis indicates that negative tail events on days when the market declines, combined with a positive relative stock-specific news sentiment CMA, are associated with improved performance in the subsequent month compared to instances with negative sentiment change. To examine this observation, this section provides To investigate this observation, an applicable trading strategy is presented in this section. Drawing on existing literature regarding market inefficiencies and the research question, this part outlines potential practical applications of news sentiment data. The strategy focuses on purchasing stocks following market declines only when accompanied by

a positive relative stock-specific news sentiment CMA (comparing 1-month to 12-month periods), which may suggest the absence of fundamental issues within the firm. In summary, the proposed reversal strategy seeks to invest for 20 trading days following tail events, conditional on the constructed relative stock-specific news sentiment metric.

4.2. Strategy specification

Aligned with the statistical analysis, the outlined trading strategy primarily targets negative stock-specific tail events. The return analysis is intended as an economic validation of the documented conditional effects rather than as an assessment of an implementable trading strategy. Investments are made in individual stocks when a company experiences a negative tail event, the market index declines, and there is a positive relative stock-specific news sentiment CMA on that particular day. In practical terms, this approach involves allocating assets to stocks that suffer a negative return jump on a day when the broader market also falls but are accompanied by an improvement in stock-specific news sentiment, suggesting an absence of underlying fundamental issues. The underlying premise is that such events prompt market overreactions, resulting in an abnormal reversal for the affected company during the subsequent month. Empirical evidence indicates that negative tail events coupled with a negative shift in news sentiment

Table 8. Strategy decomposition.

Year	Total turnover	Average number of stocks	Relative exposure stocks (%)	Relative exposure S&P 100 (%)
2004	26.75	7.72	64.04	35.96
2005	29.54	10.03	70.85	29.15
2006	27.42	9.48	65.35	34.65
2007	30.43	21.09	77.53	22.47
2008	32.02	31.91	92.74	7.26
2009	13.03	3.29	26.36	73.64
2010	18.40	12.33	48.02	51.98
2011	30.44	20.22	76.53	23.47
2012	15.99	4.75	33.87	66.13
2013	30.19	10.95	71.03	28.97
2014	38.39	16.91	81.13	18.87
2015	37.89	19.08	86.90	13.10
2016	24.10	12.94	60.70	39.30
2017	27.61	8.12	66.81	33.19
2018	37.33	28.34	89.26	10.74
2019	31.66	13.88	71.65	28.35
2020	21.59	18.95	52.64	47.36
2021	22.84	7.49	51.55	48.45

The table summarizes the strategy decomposition of the total-sample period between January 1, 2004 until December 31, 2021. It reveals the total turnover of the positive sentiment strategy throughout the years as well as the average number of stocks within the strategy and the relative exposure in stocks and in the overall S&P 100 index.

typically result in weaker post-event performance compared to those accompanied by a positive sentiment change.

This strategy is implemented across all S&P 100 constituents from 2003 to 2021, incorporating both market and news sentiment data. The backtest utilizes the 1-month over 12-month CMA as the sentiment metric, as previously discussed. Signals are processed using a two-day lag consistent with Uhl and Novacek (2021) to facilitate practical execution. Investments in targeted stocks are maintained for 20 trading days, following the methodologies established by Pritamani and Singal (2001), Chan (2003), and Savor (2012). An equal volatility-weighting scheme is applied, assigning higher allocations to lower-volatility stocks to mitigate strategy volatility associated with tail events. Due to the potential infrequency of tail event signals, the strategy may occasionally concentrate exposure in only a few companies. To prevent excessive concentration, a maximum exposure threshold of 10% per stock is enforced, promoting diversification. On days with fewer than ten signals, each selected stock receives a 10% allocation, and any remaining weight is invested in the S&P 100 index.

The strategy is strictly long-only and does not pursue short positions when negative tail events coincide with positive market days and adverse news sentiment measures. While such an approach may be suitable during periods of heightened market stress, the limited number of qualifying signals under the defined parameters—specifically the holding period and maximum exposure constraints—render a short leg impractical for the examined timeframe. Theoretically, short positions would involve selling companies

experiencing negative tail events alongside market gains and deteriorating stock-specific sentiment; however, due to the signal scarcity, this would predominantly result in shorting the S&P 100 index during normal market conditions, which is not considered viable within the context of this strategy.

4.3. Trading strategy

Based on our statistical analysis, we propose a trading strategy for the period studied that invests only in stocks with negative tail events during market downturns and positive 1-month over 12-month news sentiment CMA. The strategy frequently shifts exposure between selected stocks and the S&P 100 index. Table 8 summarizes key metrics: total turnover, average stock count, and relative exposures, showing notable variability over time.

Statistical analysis reveals a clear difference in 20-day post-event CAR between positive and negative news sentiment changes. We invest after negative tail events on poor market days when sentiment is positive, comparing results to strategies based on negative sentiment or all negative tail events. With a 5 basis point transaction cost, Figure 3 shows the cumulative NAV of such strategies from 2004–2021. Table 9 shows yearly information ratios of these strategies.

We propose that financial market mechanisms evolved throughout the entire study period, leading to possible changes in how news sentiment influences market prices. For instance, comparing the financial crisis with the Covid-19 pandemic reveals an acceleration in market dynamics, making markets more sensitive. As a result, our previous approach of using a low-frequency 1-month over 12-month news

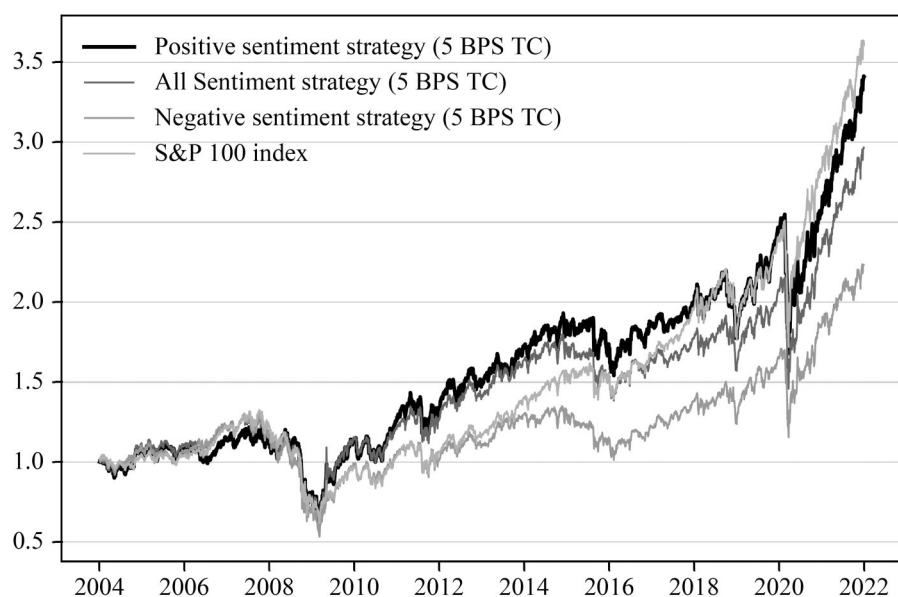


Figure 3. Strategy performance. The figure presents the NAV development (*y-axis*) of the three conditional sentiment strategies based on the 1-month over 12-month news sentiment CMA in comparison with the S&P 100 index over the in- and out-of-sample period (*x-axis*) from January 1, 2004 to December 31, 2021.

Table 9. Yearly strategy information ratios.

Year	Positive sentiment strategy (5 BPS TC)	All sentiment strategy (5 BPS TC)	Negative sentiment strategy (5 BPS TC)
2004	0.731	-0.888	-1.080
2005	0.113	0.000	-0.547
2006	-2.224	-1.418	-0.361
2007	-0.519	-1.166	-1.207
2008	0.812	0.284	-0.234
2009	0.925	0.806	0.696
2010	0.977	-0.345	-0.095
2011	0.618	0.447	-1.051
2012	0.661	0.396	-0.186
2013	-1.116	-1.007	-0.634
2014	-0.681	-1.243	-1.876
2015	-1.233	-2.415	-2.231
2016	-0.143	-0.367	-0.269
2017	-2.296	-2.324	-0.984
2018	-0.549	-0.586	-0.289
2019	0.733	-0.358	-0.140
2020	-1.517	-0.571	-0.582
2021	0.842	-0.036	-1.189

The table shows the yearly information ratios for the sentiment strategies with TC for the period between January 1, 2004 until December 31, 2021.

sentiment CMA may no longer be suitable. During the recent bull run observed out-of-sample, we recommend switching to a faster-reacting 1-week over 1-month stock-specific news sentiment CMA for greater responsiveness.

5. Conclusion

Many studies in news sentiment analysis have focused on significant jumps in stock returns. Traditionally, researchers have evaluated post-event performance by examining the presence or quality of news coverage around extreme market events. For example, Savor (2012), Boudoukh et al. (2013), and Frank and Sanati

(2018) compared shocks that were accompanied by information to those without, in order to study subsequent market reactions.

This paper introduces a new metric—a one-month over twelve-month stock-specific news sentiment crossing moving average (CMA)—which helps compare short-term versus long-term sentiment changes relative to the wider market. Using this measure, we analyze how company-specific sentiment relates to post-event outcomes.

Our empirical work, using fixed effects regression, reveals that companies with a positive news sentiment CMA tend to see notably higher cumulative abnormal returns in the month after an event, compared to

firms with negative sentiment CMAs, even after controlling for other variables. These results indicate that our sentiment measure can effectively distinguish between brief and persistent stock sell-offs. This enables the identification of profitable opportunities, such as “buying the dip” or “catching a falling knife,” within a one-month reversal framework.

The sentiment metric introduced here can be viewed as an indicator of company tone and provides valuable insights for conditional reversal trading strategies during the sample period. Our strategy outperformed the S&P 100 index, achieving an annualized Sharpe ratio of 0.27 and an information ratio of 0.36 between 2004 and 2014. However, when tested in a more dynamic market environment, this conditional sentiment strategy’s effectiveness declined. To respond to rapidly changing conditions, we propose a faster signal: a one-week over one-month sentiment CMA with a five-day holding period. By increasing signal frequency, we refine the stock-specific sentiment measure for modern markets, thus improving the agility and performance of the conditional reversal trading strategy in today’s fast-paced environment.

In summary, our findings suggest that information now moves from news media to financial markets more quickly than before. This faster pace may result from increased news reporting, a higher concentration of investors who respond to news, and improvements in automated systems that process news articles. Additionally, changes in how markets operate—particularly the growing role and influence of high-frequency trading firms—may have further strengthened this trend.

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Appendix A

Table A1. Yearly number of market tail events

Year	Positive tail events (2 std)	Negative tail events (2 std)	Positive tail events (3 std)	Negative tail events (3 std)
2003	–	–	–	–
2004	4	3	0	0
2005	5	5	1	0
2006	6	6	1	2
2007	11	23	5	10
2008	17	24	7	7
2009	2	0	0	0
2010	6	9	1	2
2011	10	16	3	4
2012	2	1	0	0
2013	3	4	1	1
2014	5	11	1	3
2015	7	12	1	3
2016	3	5	0	1
2017	4	3	0	2
2018	15	23	3	9
2019	2	5	1	0
2020	8	15	8	9
2021	2	4	0	0
Mean	5.89	8.89	1.74	2.79
Total	112	169	33	53

The table shows the yearly number of positive and negative market return jumps based on the S&P 100 index over the examined period. The tail events are based on the rolling mean and standard deviation with a look-back window of 252 trading days.

Table A2. S&P 100 constituents (full sample period).

Company name	Industry sector	PermID	Bloomberg ticker
AES Corp/The	Utilities	4295903289	AES UN Equity
APA Corp	Energy	4295903449	APA UN Equity
AT&T Corp	Communications	4295903035	653707Q UN Equity
AT&T Inc	Communications	4295904853	T UN Equity
AbbVie Inc	Consumer, Non-cyclical	5037613143	ABBV UN Equity
Abbott Laboratories	Consumer, Non-cyclical	4295903265	ABT UN Equity
Accenture PLC	Technology	4295903017	ACN UN Equity
Adobe Inc	Technology	4295905431	ADBE UW Equity
Allegheny Technologies Inc	Basic Materials	4295911994	ATI UN Equity
Allergan Ltd/Ireland	Consumer, Non-cyclical	5040050523	AGN UN Equity
Allstate Corp/The	Financial	4295903316	ALL UN Equity
Alphabet Inc	Communications	5030853586	GOOGL UW Equity
Altria Group Inc	Consumer, Non-cyclical	4295903207	MO UN Equity
Amazon.com Inc	Communications	4295905494	AMZN UW Equity
American Electric Power Co Inc	Utilities	4295903333	AEP UN Equity
American Express Co	Financial	4295903329	AXP UN Equity
American International Group Inc	Financial	4295903341	AIG UN Equity
American Tower Corp	Financial	4295903388	AMT UN Equity
Amgen Inc	Consumer, Non-cyclical	4295905537	AMGN UW Equity
Anadarko Petroleum Corp	Energy	4295903377	APC UN Equity
Anheuser-Busch Cos Inc	Consumer, Non-cyclical	4295903351	3393199Q UN Equity
Apple Inc	Technology	4295905573	AAPL UW Equity
Avon Products Inc	Consumer, Non-cyclical	4295903496	AVP UN Equity
Baker Hughes Holdings LLC	Energy	4295903515	BHI UN Equity
Bank of America Corp	Financial	8589934339	BAC UN Equity
Bank of New York Mellon Corp/The	Financial	5000001034	BK UN Equity
Bath & Body Works Inc	Consumer, Cyclical	4295904421	BBWI UN Equity
Baxter International Inc	Consumer, Non-cyclical	4295903531	BAX UN Equity
Berkshire Hathaway Inc	Financial	4295908552	BRK/B UN Equity
Biogen Inc	Consumer, Non-cyclical	4295906748	BIIB UW Equity
Black & Decker Corp/The	Industrial	4295903595	BDK UN Equity
BlackRock Inc	Financial	4295900990	BLK UN Equity
Boeing Co/The	Industrial	4295903076	BA UN Equity
Booking Holdings Inc	Communications	4295914598	BKNG UW Equity
Bristol-Myers Squibb Co	Consumer, Non-cyclical	4295903619	BMJ UN Equity
Broadcom Inc	Technology	5060689053	AVGO UW Equity
CVS Health Corp	Consumer, Non-cyclical	4295903627	CVS UN Equity
Caesars Entertainment Corp/Old	Consumer, Cyclical	4295904134	HET UN Equity
Campbell Soup Co	Consumer, Non-cyclical	4295903649	CPB UN Equity
Capital One Financial Corp	Financial	4295903652	COF UN Equity
Caterpillar Inc	Industrial	4295903678	CAT UN Equity
Celgene Corp	Consumer, Non-cyclical	4295905897	CELG UW Equity
Charter Communications Inc	Communications	4297255547	CHTR UW Equity
Chevron Corp	Energy	4295903744	CVX UN Equity
Cisco Systems Inc/Delaware	Communications	4295905952	CSCO UQ Equity
Cisco Systems Inc/Delaware	Communications	5080018615	CSCO UW Equity
Citigroup Inc	Financial	8589934316	C UN Equity
Coca-Cola Co/The	Consumer, Non-cyclical	4295903091	KO UN Equity
Colgate-Palmolive Co	Consumer, Non-cyclical	4295903113	CL UN Equity
Comcast Corp	Communications	4295908573	CMCSA UW Equity
Computer Sciences Corp	Technology	4295903765	CSC UN Equity
ConocoPhillips	Energy	4295903051	COP UN Equity
Costco Wholesale Corp	Consumer, Cyclical	4295912987	COST UN Equity
Covidien Ltd	Consumer, Non-cyclical	5000142555	COV UN Equity
Danaher Corp	Consumer, Non-cyclical	4295903833	DHR UN Equity
Dell Inc	Technology	4295906157	1683997D UW Equity
Delta Air Lines Inc/Old	Consumer, Cyclical	4295903853	DALRQ UN Equity
Devon Energy Corp	Energy	4295912097	DVN UN Equity
Dow Chemical Co/The	Basic Materials	4295903880	1697067D UN Equity
Dow Inc	Basic Materials	5000296881	DOW UN Equity
DuPont de Nemours Inc	Basic Materials	5050895314	DD UN Equity
Duke Energy Corp	Utilities	4295903896	DUK UN Equity
eBay Inc	Communications	4295906238	EBAY UW Equity
EMC Corp	Technology	4295903890	EMC UN Equity
Eastman Kodak Co	Industrial	4295903114	EKDKQ UN Equity
El Paso LLC	Energy	4295903911	EP UN Equity
Eli Lilly & Co	Consumer, Non-cyclical	4295904414	LLY UN Equity
Emerson Electric Co	Industrial	4295903920	EMR UN Equity
Entergy Corp	Utilities	4295903931	ETR UN Equity
Exelon Corp	Utilities	4295903216	EXC UN Equity
Exxon Mobil Corp	Energy	4295912121	XOM UN Equity

(Continued)

Table A2. Continued.

Company name	Industry sector	PermID	Bloomberg ticker
FedEx Corp	Industrial	4295912126	FDX UN Equity
Ford Motor Co	Consumer, Cyclical	4295903068	F UN Equity
Freeport-McMoRan Inc	Basic Materials	4295908588	FCX UN Equity
General Dynamics Corp	Industrial	4295903136	GD UN Equity
General Electric Co	Industrial	4295903128	GE UN Equity
General Motors Co	Consumer, Cyclical	4298546138	GM UN Equity
Gilead Sciences Inc	Consumer, Non-cyclical	4295906595	GILD UW Equity
Goldman Sachs Group Inc/The	Financial	4295911963	GS UN Equity
HP Inc	Technology	4295904170	HPQ UN Equity
Halliburton Co	Energy	4295904116	HAL UN Equity
Hartford Financial Services Group Inc/The	Financial	4295904141	HIG UN Equity
Hillshire Brands Co/The	Consumer, Non-cyclical	4295904880	HSH UN Equity
Home Depot Inc/The	Consumer, Cyclical	4295903148	HD UN Equity
Honeywell International Inc	Industrial	4295912155	HON UN Equity
iHeartCommunications Inc	Communications	4298056626	2968900Q UN Equity
Intel Corp	Technology	4295906830	INTC UW Equity
International Business Machines Corp	Technology	4295904307	IBM UN Equity
International Paper Co	Basic Materials	4295903177	IP UN Equity
JPMorgan Chase & Co	Financial	5000021791	JPM UN Equity
Johnson & Johnson	Consumer, Non-cyclical	4295904341	JNJ UN Equity
Kinder Morgan Inc	Energy	4298464085	KMI UN Equity
Kraft Heinz Co/The	Consumer, Non-cyclical	5038908093	KHC UW Equity
Lehman Brothers Holdings Inc	Financial	5000005634	LEHMQ UN Equity
Linde PLC	Basic Materials	5057771033	LIN UN Equity
Lockheed Martin Corp	Industrial	5000069094	LMT UN Equity
Lowe's Cos Inc	Consumer, Cyclical	4295904432	LOW UN Equity
Mastercard Inc	Financial	4295902520	MA UN Equity
McDonald's Corp	Consumer, Cyclical	4295904499	MCD UN Equity
Medimmune LLC	Consumer, Non-cyclical	4295907119	MEDI UW Equity
Medtronic PLC	Consumer, Non-cyclical	5043331632	MDT UN Equity
Merck & Co Inc	Consumer, Non-cyclical	4295904886	MRK UN Equity
MetLife Inc	Financial	4295912198	MET UN Equity
Meta Platforms Inc	Communications	4297297477	FB UW Equity
Microsoft Corp	Technology	4295907168	MSFT UW Equity
Mondelez International Inc	Consumer, Non-cyclical	4295899791	MDLZ UW Equity
Monsanto Co	Basic Materials	4295901599	1856613D UN Equity
Morgan Stanley	Financial	4295904557	MS UN Equity
NIKE Inc	Consumer, Cyclical	4295904620	NKE UN Equity
NOV Inc	Energy	4295904589	NOV UN Equity
NVIDIA Corp	Technology	4295914405	NVDA UW Equity
National Semiconductor Corp	Technology	4295904594	0203524D UN Equity
Netflix Inc	Communications	4295902158	NFLX UW Equity
NextEra Energy Inc	Utilities	4295903955	NEE UN Equity
Nokia of America Corp	Communications	4295903211	1837572D UN Equity
Norfolk Southern Corp	Industrial	4295904619	NSC UN Equity
Occidental Petroleum Corp	Energy	4295904645	OXY UN Equity
OfficeMax Inc	Consumer, Cyclical	4295903584	OMX UN Equity
Oracle Corp	Technology	4295907485	ORCL UN Equity
PayPal Holdings Inc	Consumer, Non-cyclical	5045870012	PYPL UW Equity
PepsiCo Inc	Consumer, Non-cyclical	4295904718	PEP UW Equity
Pfizer Inc	Consumer, Non-cyclical	4295904722	PFE UN Equity
Pharmacia LLC	Consumer, Non-cyclical	4296488045	748957Q UN Equity
Philip Morris International Inc	Consumer, Non-cyclical	4298015178	PM UN Equity
Procter & Gamble Co/The	Consumer, Non-cyclical	4295903247	PG UN Equity
QUALCOMM Inc	Technology	4295907706	QCOM UW Equity
RS Legacy Corp	Consumer, Cyclical	4295912262	RSHCQ UN Equity
Raytheon Co	Industrial	4295908666	RTN UN Equity
Raytheon Technologies Corp	Industrial	4295905180	RTX UN Equity
Regions Financial Corp	Financial	4295907751	RF UN Equity
Rockwell Automation Inc	Industrial	4295904837	ROK UN Equity
salesforce.com Inc	Technology	4295915633	CRM UN Equity
Schlumberger NV	Energy	4295904888	SLB UN Equity
Sears Roebuck and Co	Consumer, Cyclical	4295903054	605555Q UN Equity
Simon Property Group Inc	Financial	4295904924	SPG UN Equity
Southern Co/The	Utilities	4295903239	SO UN Equity
Sprint Communications Inc	Communications	4295904966	0848680D UN Equity
Starbucks Corp	Consumer, Cyclical	4295908005	SBUX UW Equity
T-Mobile US Inc	Communications	4295900188	TMUS UW Equity
TFCF Corp	Communications	4295903420	TFCF UW Equity
Target Corp	Consumer, Cyclical	4295912282	TGT UN Equity
Tesla Inc	Consumer, Cyclical	4297089638	TSLA UW Equity

(Continued)

Table A2. Continued.

Company name	Industry sector	PermID	Bloomberg ticker
Texas Instruments Inc	Technology	4295905063	TXN UN Equity
Thermo Fisher Scientific Inc	Consumer, Non-cyclical	4295905068	TMO UN Equity
Toys R US Inc/Old	Consumer, Cyclical	4295905112	TOY UN Equity
US Bancorp	Financial	8589934183	USB UN Equity
Union Pacific Corp	Industrial	4295905161	UNP UN Equity
Unisys Corp	Technology	4295905166	UIS UN Equity
United Parcel Service Inc	Industrial	4295912318	UPS UN Equity
UnitedHealth Group Inc	Consumer, Non-cyclical	5046300101	UNH UN Equity
Verizon Communications Inc	Communications	4295911976	VZ UN Equity
ViacomCBS Inc	Communications	4295908715	VIAC UN Equity
Visa Inc	Financial	4298015179	V UN Equity
Walgreens Boots Alliance Inc	Consumer, Cyclical	5043951500	WBA UN Equity
Walmart Inc	Consumer, Cyclical	4295905298	WMT UN Equity
Walt Disney Co/The	Communications	5064610769	DIS UN Equity
Wells Fargo & Co	Financial	8589934175	WFC UN Equity
Weyerhaeuser Co	Financial	4295903255	WY UN Equity
Williams Cos Inc/The	Energy	4295905343	WMB UN Equity
Wyeth LLC	Consumer, Non-cyclical	4295903346	WYE UN Equity
3M Co	Industrial	5000072036	MMM UN Equity